MD FAHIM FAYSAL

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Rajshahi, Bangladesh

WORK EXPERIENCE

Research Assistant | Bioinformatics Lab (Dry), Department of Statistics, Rajshahi University February, 2024 - Present

- Designed and implemented bioinformatics algorithms, R packages, and web-based platforms for drug and vaccine discovery through integrative analysis of molecular OMICS data, including genomics, transcriptomics, proteomics, and metagenomics.

RemoteIntegrity LLC, Clearwater, Florida, USA | *Software Developer* **January 2025-July 2025**

- Developed a custom CRM system with Node.js, MySQL, and React.js, integrating APIs with platforms like Apploye and YouTube, and built a scalable job application and course management platform attracting over 1000 daily users.

COMPETITIVE PROGRAMMING EXPERIENCES

- **Expert** at **Codeforces Max. Rating 1686**
- Solved 650+ problems in Codeforces
- **Max. Rating 1744** at CodeChef
- Solved 200+ problems in different online judges like Atcoder, Hacherearth, LightOj, LeetCode, etc

EDUCATION

Rajshahi University of Engineering & Technology	Feb 2019 - May 2025
Bachelor of Science in Computer Science & Engineering	CGPA 3.31/4.00
Last 4 semesters average	CGPA 3.50/4.00
Adamjee Cantonment College	July 2016 - April 2018
Higher Secondary School Certificate	**GPA 5.00/5.00**

PUBLICATIONS

Published Articles

- Ahmmed, R., Antu, U.S., Noor, T., **Faysal, M. F.**, Akter, M.T., Nesa, M., Mollah, M.N.H. Discovery of shared molecular signatures and their functions associated with type-2 diabetes and myocardial infarction, and repurposing common drugs. Comput Biol Chem. 2026 Mar 27;123:109036. doi: 10.1016/j.compbiolchem.2026.109036.
- Ahmmed, R., Antu, U.S., Noor, T., **Faysal, M. F.**, Sarker, A., Mahmud, S., Ajadee, A., Mollah, M.N.H. Genetic underpinnings of type-2 diabetes (T2D) with colorectal cancer (CRC): In-silico discovery of common molecular signatures, pathogenetic processes and therapeutic candidates. J Genet Eng Biotechnol. 2026 Mar;24(1):100667. doi: 10.1016/j.jgeb.2026.100667.
- Ali, M., Ahsan, M. A., Islam, M. S., **Faysal, M. F.**, Ryan, H. M., Sipa, U. T. A., Hossain, M. S., Kabir, M. H., Hasan, M. M., Pal, N. K., & Mollah, M. N. H. In-silico identification of SNPs associated with breast cancer for disclosing pathogenetic processes and therapeutic candidates. Discov Onc (2026). https://doi.org/10.1007/s12672-026-05809-0
- Ahmed, M. F., Noman, M. A., Ahmed, M. F., Latif, M. A., Pappu, M. a. A., Islam, M. S., Hossain, M. S., Hossen, M. B.,**Faysal, M. F.**, Hasan, M. M., & Mollah, M. N. H. (2025). In-Silico discovery of Pediatric Acute-Myeloid-Leukemia (pAML) causing druggable molecular signatures highlighting their pathogenetic processes and therapeutic agents through single-cell RNA-Seq profile analysis. *PLOS ONE*, 20(10), e0335410. https://doi.org/10.1371/journal.pone.0335410

- Pappu, M. A. A., Alamin, M., Noman, M. A., Sultana, M. H., Ahmed, M. F., Hossain, M. S., Latif, M. A., **Faysal, M. F.**, Azad, A., Alyami, S. A., Alotaibi, N., & Mollah, M. N. H. (2025). Multi-Transcriptome-Informed Network Pharmacology Reveals Novel Biomarkers and Therapeutic Candidates for Parkinson's Disease. Genes, 16(12), 1459. https://doi.org/10.3390/genes16121459
- Islam, M. S., Mollah, M. M. H., Ahsan, M. A., Ali, M., Noman, M. A., Ahmed, M. F., Latif, M. A., Pal, N. K., **Faysal, M. F.**, & Mollah, M. N. H.(2026).In-silico identification of genetic variants associated with chronic lymphocytic leukemia for diagnostic and therapeutic applications. Sci Rep 16, 14545.https://doi.org/10.1038/s41598-026-45456-7

Under Review

- **Faysal, M. F.**., Noor, T., Shyla, S., Sharma, O., Ahmed, M. F., Ahmed, R., Kabir, M. M. J., & Mollah, M. N. H. Bioinformatics-guided discovery of common genetic underpinnings through which idiopathic pulmonary fibrosis and tuberculosis stimulate each other, and repurposing common therapies. *Genes* **[Under Review]**
- **Faysal, M. F.**, Noor, T., Shyla, S., Noor, A., Ahmed, M. F., Sharma, O., Kabir, M. M. J., & Mollah, M. N. H. In-silico discovery of druggable common hub-genes and associated factors through which type-2 diabetes stimulates tuberculosis, and repurposing common drugs through integrated RNA-seq profile analysis. Infection, Genetics and Evaluation **[Under Review]**
- Noor, T., **Faysal, M. F.**, Ahmed, R., Ahmed, M. F., Sharma, O., Kabir, M. H., Kabir, M. R., Kabir, M. M. J., & Mollah, M. N. H. Computational discovery of common host key-genes (chKGs) and their mechanisms associated with Dengue-SARS-CoV-2 co-infection for exploring common therapies through integrated RNA-seq profile analysis. *Viruses* **[Under Review]**

Preprints

- Ahmed, M. F., **Faysal, M. F.**, Sawad, K. M., Elahi, M. A. E., Noor, T., Kibria, M. K., Hasan, M. M., & Mollah, M. N. H. (2026). *FlexAutoDock: A flexible platform for automated molecular docking and virtual screening of natural and synthetic compounds*. bioRxiv. https://doi.org/10.64898/2026.08.11.744098
- Sharma, O., Ahmed, M. F., Sharma, D., Sharma, A., Noor, T., **Faysal, M. F.**, Ahmed, M. F., Hossain, M. S., Noman, M. A., Latif, M. A., Ali, M., Ahmed, D. M., & Mollah, M. N. H. (2026). *In silico transcriptomic analysis reveals shared molecular signatures and immune-associated pathways between Hashimoto's thyroiditis and type 2 diabetes with exploratory drug repurposing*. bioRxiv. https://doi.org/10.64898/2026.02.16.706089

Conference Proceedings

- **Faysal, M. F.**, Afroge, S., & Noor, T.. (2024, November). In-Silico Identification of Shared Hub-Genes Between Idiopathic Pulmonary Fibrosis and Tuberculosis Diseases & Drug Repurposing. In 2024 IEEE International Conference on Biomedical Engineering, Computer and Information Technology for Health (BECITHCON) (pp. 67-72). IEEE.
- Noor, T.,**Faysal, M. F.**, Sadad, N., & Mondal, M. N. I. (2024, December). Discovery of SARS-COV-2 and Dengue-Virus Co-Infections Causing Shared Host Key-Genes using Bioinformatics and Machine Learning Approaches for Diagnosis and Therapies. In Proceedings of the 8th International Conference on the Role of Statistics and Data Science in 4IR (ICRSDS4IR). Rajshahi, Bangladesh.

TECHNICAL SKILLS

- **Programming & Scripting** C, C++, Java, JavaScript (Node.js, React.js, Angular.js), Python, R, HTML, CSS, SQL (MySQL, PostgreSQL)
- **Machine Learning/Deep Learning:** NumPy, Pandas, Matplotlib, OpenCV, Scikit-learn,Random Forest
- **Bioinformatics & Computational Biology:** LIMMA, DESeq2, WCGNA, PPI Network Analysis, Data Visualization (ggplot2, ComplexHeatmap, Plotly), Biomarker Discovery, Molecular Docking (AutoDock Vina), ADMET Analysis (SwissADME, pkCSM, ADMETlab), Drug Repurposing Pipelines
- **Cheminformatics & Structural Biology:** RDKit, Open Babel, PyMOL, Discovery Studio Visualizer.

AWARDS

- Leadership Award Class Representative 2019-2021